Cross-agency data matching

One identity across many programs

Status: Draft v0.1 (2026-11-04), for Steve's review. Extension 1 of 10 (see the extensions overview). Reference code: `analytics/forensic/crossmatch.py`, tested on synthetic data only.

Bottom line

Most large frauds are invisible inside one program and obvious across several: the same person drawing unemployment insurance in two states and a PPP loan that claims them as a full-time employee; a hospice owner who was debarred by GSA and is still billing Medicare; a dead veteran's identity cashing SNAP benefits. No agency can see this alone, because each holds one slice of the identity. This extension builds the shared method: link people and businesses across program files with privacy-preserving keys, then run four checks that only a linked view allows.

1. **Stacking and exclusive programs:** one identity in more programs than its profile explains, or in two programs that the law says cannot both pay for the same thing at the same time.
2. **Paid after death:** a death recorded in any one program reaches every program the identity is paid by.
3. **Paid while excluded:** SAM exclusions, the HHS-OIG LEIE and OPM debarments applied to every payment stream, not only the one that issued the exclusion.
4. **One identifier, many identities:** one bank account, phone, email or device used by people or firms that otherwise do not link (roadmap idea X-2).

Phase 1 runs the business side on public data (SAM.gov, USAspending, SBA loan-level files, CMS enrollment files, exclusion lists). Phase 2 adds person-level data through Do Not Pay and computer matching agreements (see the policy and legal track).

1. Why cross-agency matching is the highest-value extension

- The roadmap's six cross-agency ideas (X-1 to X-6) all depend on it.
- GAO and the Pandemic Response Accountability Committee found the same identities across PPP, EIDL and unemployment insurance; the pandemic analytics center that matched them closed when its authority lapsed.
- Treasury's Do Not Pay already matches payments against death and exclusion lists, but only for agencies that send it files, and only on the identifiers they send. Matching on linked identities catches the variants that exact matching misses.

2. Data sources

Source	Entities	Keys	Phase
SAM.gov entity registrations and exclusions	Businesses, individuals excluded	UEI, CAGE, name, address	1 (public)
USAspending and FPDS	Contract, grant and loan recipients	UEI, name, address	1
SBA PPP and EIDL loan-level files	Borrowers	Name, address, lender, NAICS	1
CMS PECOS, NPPES, provider enrollment files	Providers and owners	NPI, name, address, owners	1
HHS-OIG LEIE, OPM debarments, state Medicaid exclusions	Excluded people and firms	NPI, name, DOB (LEIE), address	1
Treasury payment files (through Do Not Pay)	Payees	TIN, bank routing and account	2
SSA death data (full file through Do Not Pay)	People	SSN, name, DOB	2
State UI, SNAP and Medicaid eligibility files	Claimants	SSN, name, DOB, address, bank, device	2 (state agreements)
NASWA Integrity Data Hub results	UI claims	Cross-state flags	2

3. Method

Normalize every name and address the same way: legal suffixes and punctuation removed, USPS standardization (CASS-certified in production), phones to E.164, emails lowercased with plus-tags removed.

Hash identifiers inside each owner's boundary. TINs, SSNs and bank accounts never leave the agency that holds them. Each data owner computes a keyed hash (HMAC-SHA256) of the normalized value with a key held by the matching authority, and only hashes are exchanged. Two equal hashes mean two records share an identifier without revealing it. Keys rotate per matching agreement.

Block, then compare. Candidate pairs are records that share a hashed TIN, bank account, phone or email, or a first name token and ZIP. Blocks larger than 200 records (registered agents, call centers) are dropped, as hubs are in the network method. Each pair is scored with the Fellegi-Sunter weights already in the forensic methods handbook section 7.

Link and review. Pairs above the upper threshold link automatically; pairs between the thresholds go to clerical review. Thresholds are set on 500 hand-labeled pairs to reach 95% precision, as the methodology requires.

Run the checks in section 4 on every linked identity.

4. The four checks



Check	Rule	Grace and exceptions	Typical cause
Stacking	Identity in more than three programs, or in a pair listed as exclusive	Written per pair from program law (for example UI and full-time payroll in the same weeks; two state SNAP cases in the same month)	Identity theft, eligibility fraud
Paid after death	Any payment dated more than the grace period after the earliest death date on any linked record	Month-of-death payments; claims for services before death	Identity theft, family members keeping benefits
Paid while excluded	Payment dated after an exclusion start in any list	Services before exclusion; reinstatements	Excluded providers working through new entities
Shared identifier	One bank account, phone, email or device used by three or more identities that did not link	Payroll processors, tax preparers, shelters, care homes (benign-hub list)	Mule accounts, synthetic identities, ghost employees

A flag is a reason to look, never a finding. Each one carries the linked records and the reason, and goes to the owning program or OIG under the lead package format in the review plan.

5. Privacy and safeguards

- **Legal basis first.** Person-level matching across agencies is a computer matching program under the Privacy Act unless an exception applies (OIG investigations, Do Not Pay under the Payment Integrity Information Act). See the policy and legal track.
- **Minimum necessary.** Only the fields in section 3 cross a boundary, and only as hashes where they are identifiers.
- **Due process.** No payment is stopped on a match alone; the Privacy Act requires independent verification and notice before adverse action.
- **Error budget.** Each check reports its false-positive rate from the clerical review sample, and checks above 20% false positives are not released.
- **Bias review.** Match rates by state and by name origin are compared, since name-based matching is less accurate for some naming conventions.

6. Phase 1 deliverable (public data, weeks 5 to 9)

A business-side linked file across SAM.gov, USAspending, SBA loan files and CMS enrollment, with three outputs:

1. Excluded parties (SAM, LEIE) linked to entities still receiving awards or loans after the exclusion date.
2. Entities in four or more programs with growth and address signals.
3. Bank-free “shared identifier” clusters using phone, email and address from public registrations.

7. Phase 2 deliverable



A Do Not Pay use case for one cross-program check (paid after death across FEHB, VA and SNAP, roadmap X-1), and one state pair for UI and SNAP identity reuse (roadmap X-3), under the pilot kit.

8. Questions for Steve

1. Does the SBA work with Rosemary already hold linked identities we can reuse as the seed and labeled set?
2. Lead the Phase 2 ask with Treasury Do Not Pay (faster, narrower) or with one OIG (broader, slower)?

